Ultimate DevOps + AWS Engineer Guide

====================================

Index:

----

1. K3s - Lightweight Kubernetes Distribution

Explanation:

K3s is a lightweight Kubernetes distribution designed for IoT and edge computing. It's easy to install, uses less memory and CPU, and is suitable for development, testing, and small production workloads.

Workflow:

- Install K3s

- Deploy services using Kubernetes YAML files

- Use kubectl to manage the cluster

- Monitor logs and resource usage

Real-time Scenario Q&A:

Q1: How do you install K3s on a Linux VM?

A1: Run `curl -sfL https://get.k3s.io | sh -` on the host system.

Q2: How do you access the K3s Kubernetes dashboard?

A2: Install it using Helm or Kubernetes manifests, then port-forward to view it.

... (Continue with 25 Q&A for K3s)

----

2. Argo CD - GitOps for Kubernetes

Explanation:

Argo CD is a declarative, GitOps continuous delivery tool for Kubernetes. It tracks application states in Git and automatically syncs changes.

Workflow:

- Install Argo CD

- Connect to Git repo

- Deploy applications via Git manifests or Helm charts

- Monitor synchronization and health

Real-time Scenario Q&A:

Q1: What is GitOps and how does Argo CD implement it?

A1: GitOps is using Git as the single source of truth. Argo CD monitors the Git repository and applies the changes to the Kubernetes cluster.

... (Continue with 25 Q&A for Argo CD)

(Sections for Grafana Loki, HashiCorp Vault, Helm will follow a similar structure)